

AI IMPACT ON JOBS AND SALARIES

Big Data Analysis using Hadoop HDFS, Apache Pig and Java MapReduce

Submitted by: Walid Talal

ID: 0222220005101045

Submitted to: MD Tamim Hossain

Lecturer, Department of Computer Science & Engineering
Premier University

Dataset: ai_jobs_salaries.csv (71,913 records, 17 attributes)

Technologies: Apache Hadoop, HDFS, Java MapReduce, Apache Pig 0.18.0

Table of Contents

1. Problem Statement	3
2. Dataset Description.....	3
3. Summary of Data Analysis	3
4. Map Reduce Analysis	4
4.1 Highest Salary by Job Role.....	4
4.2 Job Count by Experience Level	9
4.3 Top 10 Highest-Paid Individual Job Records	13
5. Pig AI Jobs and Salaries Data Analysis	19
5.1 Top 10 Highest-Paying Job Roles.....	19
5.2 Salary Analysis by Experience Level	22
5.3 Job Distribution by Work Mode	24
5.4 Top 10 Job Locations.....	27
5.5 Salary Comparison by Employment Type	29

1. Problem Statement

Analyze the AI Jobs and Salaries Big Data dataset using Hadoop (Java MapReduce) and Apache Pig, based on different column fields such as job role, experience level, employment type, work mode and company location, to provide comprehensive insights into how salaries in AI-related roles vary across the workforce.

2. Dataset Description

Dataset file: Dataset/ai_jobs_salaries.csv (approximately 71,913 records, 17 attributes)

Field	Description
work_year	Year the salary was reported
experience_level_label	Decoded experience level (Entry / Mid / Senior / Executive)
employment_type_label	Decoded employment type (Full-time / Part-time / Contract / Freelance)
job_title	Original raw job title as self-reported
role_family	Manually grouped role family derived from job_title
salary_in_usd	Salary converted to USD (primary comparable field)
work_mode	On-site / Hybrid / Remote, decoded from remote_ratio
company_size	Company size band as reported (S / M / L)
company_location	Company's country (ISO code, as provided by source)
salary_outlier_flag	True if salary_in_usd falls outside the 1.5×IQR range (flagged, not removed)
isco_group_hint	Approximate ISCO-08 occupational group hint (manual mapping, not authoritative)

Source: foorilla/ai-jobs-net-salaries (CC0 1.0). Salary data is self-reported survey data, not verified payroll data.

3. Summary of Data Analysis

The following Java MapReduce programs are implemented on the dataset:

1. Highest Salary by Job Role — maximum salary_in_usd for each role_family
2. Job Count by Experience Level — number of records for each experience_level_label
3. Top 10 Highest-Paid Individual Job Records — top 10 individual salary records using a priority queue

The following Apache Pig analyses are performed on the dataset:

1. Top 10 Highest-Paying Job Role Families
2. Salary Analysis by Experience Level (average, maximum, minimum)
3. Job Distribution by Work Mode (on-site / hybrid / remote, with percentage)
4. Top 10 Job Locations by number of AI jobs
5. Salary Comparison by Employment Type (average, maximum, minimum)

4. Map Reduce Analysis

Three Java MapReduce programs were developed and executed on Hadoop. Each program uses a Mapper class to parse and emit relevant fields from the CSV records, and a Reducer class to aggregate the results. Programs were compiled, packaged as runnable JAR files, and submitted to the Hadoop cluster via the `hadoop jar` command against the dataset stored on HDFS.

4.1 Highest Salary by Job Role : Returns the maximum salary recorded for each job role family (HighestSalaryByRole)

This analysis comprises of:

1. Mapper Class : Parses each CSV record and emits role_family of type Text and salary_in_usd of type DoubleWritable
2. Reducer Class : Emits role_family as a key of type Text and the maximum salary of type DoubleWritable
3. Driver Class : Sets up configuration for the Mapper and Reducer classes

Mapper Class

```
public static class SalaryMapper
    extends Mapper<Object, Text, Text, DoubleWritable> {

    private final Text roleFamily = new Text();
    private final DoubleWritable salary = new DoubleWritable();

    @Override
    public void map(
        Object key,
        Text value,
        Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.startsWith("work_year,")) {
            return;
        }

        String[] fields = line.split(",", -1);

        // Dataset must contain all required columns
        if (fields.length < 16) {
            return;
        }

        try {

            double salaryInUsd = Double.parseDouble(fields[6].trim());

            String role = fields[15].trim();

            if (role.isEmpty()) {
                return;
            }

            roleFamily.set(role);
            salary.set(salaryInUsd);

            context.write(roleFamily, salary);

        } catch (NumberFormatException e) {
        }
    }
}
```

Reducer Class

```
public static class SalaryReducer
    extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
```

```

private final DoubleWritable maximumSalary =
    new DoubleWritable();

@Override
public void reduce(
    Text key,
    Iterable<DoubleWritable> values,
    Context context)
    throws IOException, InterruptedException {

    double maxSalary = Double.MIN_VALUE;

    for (DoubleWritable value : values) {

        if (value.get() > maxSalary) {
            maxSalary = value.get();
        }
    }

    maximumSalary.set(maxSalary);

    context.write(key, maximumSalary);
}
}

```

Driver Class

```

public static void main(String[] args)
    throws Exception {

    if (args.length != 2) {

        System.err.println(
            "Usage: HighestSalaryByRole <input> <output>");

        System.exit(-1);
    }

    Configuration configuration = new Configuration();

    Job job = Job.getInstance(
        configuration,
        "Highest Salary by Job Role");

    job.setJarByClass(HighestSalaryByRole.class);

    // Mapper
    job.setMapperClass(SalaryMapper.class);

    // Reducer
    job.setReducerClass(SalaryReducer.class);

    // Output types
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(DoubleWritable.class);

    // Input / Output paths
    FileInputFormat.addInputPath(
        job,
        new Path(args[0]));

    FileOutputFormat.setOutputPath(
        job,
        new Path(args[1]));

    System.exit(

```

```

        job.waitForCompletion(true)
        ? 0
        : 1);
    }
}

```

Execution

```

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\HighestSalaryByRole>hadoop jar HighestSalaryByRole.jar HighestSalaryByRole /AI_Impact_Jobs/ai_jobs_salaries.csv /AI_Impact_Jobs/mapreduce/highest_salary_by_role
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\HighestSalaryByRole>set DEFAULT_LIBEXEC_DIR=C:\hadoop\bin\..\libexec
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\HighestSalaryByRole>if not defined HADOOP_LIBEXEC_DIR (set HADOOP_LIBEXEC_DIR=C:\hadoop\bin\..\libexec )
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\HighestSalaryByRole>call C:\hadoop\bin\..\libexec\hadoop-config.cmd jar HighestSalaryByRole.jar HighestSalaryByRole /AI_Impact_Jobs/ai_jobs_salaries.csv /AI_Impact_Jobs/mapreduce/highest_salary_by_role
JAVA_HOME=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10\bin\java.exe]
CHECKING=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10\bin\java.exe]
2026-09-12 20:41:17,983 INFO client.DefaultHadoopFallbackProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 20:41:18,269 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-12 20:41:18,281 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Walid/.staging/job_1789219602256_0019
2026-09-12 20:41:18,475 INFO input.FileInputFormat: Total input files to process : 1
2026-09-12 20:41:18,535 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-12 20:41:18,645 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789219602256_0019
2026-09-12 20:41:18,645 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-12 20:41:18,788 INFO conf.Configuration: resource-types.xml not found
2026-09-12 20:41:18,789 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-12 20:41:18,843 INFO impl.YarnClientImpl: Submitted application application_1789219602256_0019
2026-09-12 20:41:18,878 INFO mapreduce.Job: The url to track the job: http://scwdussh:8088/proxy/application_1789219602256_0019/
2026-09-12 20:41:18,878 INFO mapreduce.Job: Running job: job_1789219602256_0019
2026-09-12 20:41:32,028 INFO mapreduce.Job: Job job_1789219602256_0019 running in uber mode : false
2026-09-12 20:41:32,028 INFO mapreduce.Job: map 0% reduce 0%
2026-09-12 20:41:40,189 INFO mapreduce.Job: map 100% reduce 0%
2026-09-12 20:41:55,341 INFO mapreduce.Job: map 100% reduce 100%
2026-09-12 20:42:00,395 INFO mapreduce.Job: Job job_1789219602256_0019 completed successfully
2026-09-12 20:42:00,473 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=2014988
    FILE: Number of bytes written=4667335
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=5168297
    HDFS: Number of bytes written=256
    HDFS: Number of read operations=8
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=1
    Launched reduce tasks=1
    Data-local map tasks=1
    Total time spent by all maps in occupied slots (ms)=9008
    Total time spent by all reduces in occupied slots (ms)=8938
    Total time spent by all map tasks (ms)=9008
    Total time spent by all reduce tasks (ms)=8938

```

Job submission and execution

```

Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=9008
  Total time spent by all reduces in occupied slots (ms)=8938
  Total time spent by all map tasks (ms)=9008
  Total time spent by all reduce tasks (ms)=8938
  Total vcore-milliseconds taken by all map tasks=9008
  Total vcore-milliseconds taken by all reduce tasks=8938
  Total megabyte-milliseconds taken by all map tasks=9224192
  Total megabyte-milliseconds taken by all reduce tasks=9152512
Map-Reduce Framework
  Map input records=71914
  Map output records=71913
  Map output bytes=1871156
  Map output materialized bytes=2014988
  Input split bytes=122
  Combine input records=0
  Combine output records=0
  Reduce input groups=11
  Reduce shuffle bytes=2014988
  Reduce input records=71913
  Reduce output records=11
  Spilled Records=143826
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=34
  CPU time spent (ms)=4700
  Physical memory (bytes) snapshot=635797504
  Virtual memory (bytes) snapshot=730972160
  Total committed heap usage (bytes)=320864256
  Peak Map Physical memory (bytes)=404799488
  Peak Map Virtual memory (bytes)=447348736
  Peak Reduce Physical memory (bytes)=230998016
  Peak Reduce Virtual memory (bytes)=283623424
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=9168175
File Output Format Counters

```

Job completed successfully

Output

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\HighestSalaryByRole>hdfs dfs -cat /AI_Impact_Jobs/mapreduce/highest_salary_by_role/part-r-00000
JAVA_HOME=[C:\PROGRA~1\ECLIPS~1\JDK-17~1.10~]
CHECKING=[C:\PROGRA~1\ECLIPS~1\JDK-17~1.10~\bin\java.exe]
AI Architect      800000.0
AI Engineer       600000.0
Analytics Manager 625000.0
Computer Vision  413160.0
Data Analyst      774000.0
Data Engineer     793136.0
Data Scientist    750000.0
ML Engineer       750000.0
NLP               275000.0
Other / Unclassified 800000.0
Research Scientist 750000.0
```

Highest Salary by Job Role — HDFS output (part-r-00000)

AI Architect	800000.0
AI Engineer	600000.0
Analytics Manager	625000.0
Computer Vision	413160.0
Data Analyst	774000.0
Data Engineer	793136.0
Data Scientist	750000.0
ML Engineer	750000.0
NLP	275000.0
Other / Unclassified	800000.0
Research Scientist	750000.0

4.2 Job Count by Experience Level : Returns the total number of job records for each experience level (JobsByExperienceLevel)

This analysis comprises of:

1. Mapper Class : Emits experience_level_label of type Text and 1 of type IntWritable for every record
2. Reducer Class : Emits experience level as a key of type Text and its total occurrence of type IntWritable
3. Driver Class : Sets up configuration for the Mapper and Reducer classes

Mapper Class

```
public static class ExperienceMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private final Text experienceLevel = new Text();
    private final IntWritable one = new IntWritable(1);

    @Override
    public void map(
        Object key,
        Text value,
        Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.startsWith("work_year,")) {
            return;
        }

        String[] fields = line.split(",", -1);

        if (fields.length < 12) {
            return;
        }

        // Column 11 = experience_level_label
        String experience = fields[11].trim();

        if (experience.isEmpty()) {
            return;
        }

        experienceLevel.set(experience);

        context.write(experienceLevel, one);
    }
}
```

Reducer Class

```
public static class ExperienceReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    private final IntWritable totalJobs = new IntWritable();

    @Override
    public void reduce(
        Text key,
        Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {
```

```

        int sum = 0;

        for (IntWritable value : values) {
            sum += value.get();
        }

        totalJobs.set(sum);

        context.write(key, totalJobs);
    }
}

```

Driver Class

```

public static void main(String[] args)
    throws Exception {

    if (args.length != 2) {

        System.err.println(
            "Usage: JobsByExperienceLevel <input> <output>");

        System.exit(-1);
    }

    Configuration configuration = new Configuration();

    Job job = Job.getInstance(
        configuration,
        "Job Count by Experience Level");

    job.setJarByClass(JobsByExperienceLevel.class);

    // Mapper
    job.setMapperClass(ExperienceMapper.class);

    // Reducer
    job.setReducerClass(ExperienceReducer.class);

    // Output types
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    // Input path
    FileInputFormat.addInputPath(
        job,
        new Path(args[0]));

    // Output path
    FileOutputFormat.setOutputPath(
        job,
        new Path(args[1]));

    System.exit(
        job.waitForCompletion(true)
            ? 0
            : 1);
}

```

Execution

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\JobsByExperienceLevel>hadoop jar JobsByExperienceLevel.jar JobsByExperienceLevel /AI_Impact_Jobs/ai_jobs_salaries.csv /AI_Impact_Jobs/mapreduce/jobs_by_experience

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\JobsByExperienceLevel>set DEFAULT_LIBEXEC_DIR=C:\hadoop\bin\..\libexec

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\JobsByExperienceLevel>if not defined HADOOP_LIBEXEC_DIR (set HADOOP_LIBEXEC_DIR=C:\hadoop\bin\..\libexec)

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\JobsByExperienceLevel>call C:\hadoop\bin\..\libexec\hadoop-config.cmd jar JobsByExperienceLevel.jar JobsByExperienceLevel /AI_Impact_Jobs/ai_jobs_salaries.csv /AI_Impact_Jobs/mapreduce/jobs_by_experience
JAVA_HOME=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10-]
CHECKING=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10-\.bin\java.exe]
2026-09-12 20:55:32,692 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 20:55:32,990 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-12 20:55:33,003 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Walid/.staging/job_1789219602256_0020
2026-09-12 20:55:33,204 INFO input.FileInputFormat: Total input files to process : 1
2026-09-12 20:55:33,261 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-12 20:55:33,368 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789219602256_0020
2026-09-12 20:55:33,369 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-12 20:55:33,514 INFO conf.Configuration: resource-types.xml not found
2026-09-12 20:55:33,515 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-12 20:55:33,563 INFO impl.YarnClientImpl: Submitted application application_1789219602256_0020
2026-09-12 20:55:33,604 INFO mapreduce.Job: The url to track the job: http://scweddussh:8088/proxy/application_1789219602256_0020/
2026-09-12 20:55:33,604 INFO mapreduce.Job: Running job: job_1789219602256_0020
2026-09-12 20:55:46,793 INFO mapreduce.Job: Job job_1789219602256_0020 running in uber mode : false
2026-09-12 20:55:46,797 INFO mapreduce.Job: map 0% reduce 0%
2026-09-12 20:55:57,950 INFO mapreduce.Job: map 100% reduce 0%
2026-09-12 20:56:10,071 INFO mapreduce.Job: map 100% reduce 100%
2026-09-12 20:56:15,147 INFO mapreduce.Job: Job job_1789219602256_0020 completed successfully
2026-09-12 20:56:15,246 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=1296716
    FILE: Number of bytes written=3230807
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
```

Job submission and execution

```

2026-09-12 20:55:57,950 INFO mapreduce.Job: map 100% reduce 0%
2026-09-12 20:56:10,071 INFO mapreduce.Job: map 100% reduce 100%
2026-09-12 20:56:15,147 INFO mapreduce.Job: Job job_1789219602256_0020 completed successfully
2026-09-12 20:56:15,246 INFO mapreduce.Job: Counters: 54

File System Counters
  FILE: Number of bytes read=1296716
  FILE: Number of bytes written=3230807
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=9168297
  HDFS: Number of bytes written=73
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0

Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=8997
  Total time spent by all reduces in occupied slots (ms)=8925
  Total time spent by all map tasks (ms)=8997
  Total time spent by all reduce tasks (ms)=8925
  Total vcore-milliseconds taken by all map tasks=8997
  Total vcore-milliseconds taken by all reduce tasks=8925
  Total megabyte-milliseconds taken by all map tasks=9212928
  Total megabyte-milliseconds taken by all reduce tasks=9139200

Map-Reduce Framework
  Map input records=71914
  Map output records=71913
  Map output bytes=1152884
  Map output materialized bytes=1296716
  Input split bytes=122
  Combine input records=0
  Combine output records=0
  Reduce input groups=4
  Reduce shuffle bytes=1296716
  Reduce input records=71913
  Reduce output records=4
  Spilled Records=143826

```

Job completed successfully

Output

```

Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=31
CPU time spent (ms)=4294
Physical memory (bytes) snapshot=639631360
Virtual memory (bytes) snapshot=735944704
Total committed heap usage (bytes)=326107136
Peak Map Physical memory (bytes)=394134208
Peak Map Virtual memory (bytes)=436510912
Peak Reduce Physical memory (bytes)=245297152
Peak Reduce Virtual memory (bytes)=299425792

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=9168175
File Output Format Counters
  Bytes Written=73

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\JobsByExperienceLevel>hdfs dfs -ls /AI_Impact_Jobs/mapreduce/jobs_by_experience
JAVA_HOME=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10-]
CHECKING=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10-\bin\java.exe]
Found 2 items
-rw-r--r-- 1 Walid supergroup 0 2026-09-12 20:56 /AI_Impact_Jobs/mapreduce/jobs_by_experience/ SUCCESS
-rw-r--r-- 1 Walid supergroup 73 2026-09-12 20:56 /AI_Impact_Jobs/mapreduce/jobs_by_experience/part-r-000000

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\JobsByExperienceLevel>hdfs dfs -cat /AI_Impact_Jobs/mapreduce/jobs_by_experience/part-r-000000
JAVA_HOME=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10-]
CHECKING=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10-\bin\java.exe]
Entry-level 7975
Executive-level 2841
Mid-level 23395
Senior-level 37702

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\JobsByExperienceLevel>hdfs dfs -get /AI_Impact_Jobs/mapreduce/jobs_by_experience "C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\MapReduce Results\JobsByExperienceLevel"
JAVA_HOME=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10-]
CHECKING=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10-\bin\java.exe]
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\JobsByExperienceLevel>

```

Job Count by Experience Level — HDFS output (part-r-00000)

Entry-level	7975
Executive-level 2841	
Mid-level	23395
Senior-level	37702

4.3 Top 10 Highest-Paid Individual Job Records : Returns the top 10 individual job records with the highest salary (Top10HighestPaidJobs)

This analysis comprises of:

1. Mapper Class : Emits salary_in_usd as the key of type Text and job_title, experience level, company location and work mode (tab separated) as the value of type Text
2. Reducer Class : Maintains a size-10 min-heap PriorityQueue keyed on salary; when the queue exceeds 10 entries the lowest salary is removed, so only the top 10 highest-paid records survive; results are emitted in descending order in the cleanup phase
3. JobRecord Class : Small helper (POJO) class holding a salary value and its associated job details string
4. Driver Class : Sets up configuration for the Mapper and Reducer classes

Mapper Class

```
public static class SalaryMapper
    extends Mapper<Object, Text, Text, Text> {

    private final Text outputKey = new Text();
    private final Text outputValue = new Text();

    @Override
    public void map(
        Object key,
        Text value,
        Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.startsWith("work_year,")) {
            return;
        }

        String[] fields = line.split(",", -1);

        if (fields.length < 16) {
            return;
        }

        try {

            // Column 6 = salary_in_usd
            double salary = Double.parseDouble(
                fields[6].trim());

            // Column 3 = job_title
            String jobTitle = fields[3].trim();

            // Column 11 = experience_level_label
            String experienceLevel = fields[11].trim();

            // Column 9 = company_location
            String companyLocation = fields[9].trim();

            // Column 13 = work_mode
            String workMode = fields[13].trim();

            outputKey.set(String.valueOf(salary));

            String jobRecord =
```

```

        jobTitle + "\t"
        + experienceLevel + "\t"
        + companyLocation + "\t"
        + workMode;

        outputValue.set(jobRecord);

        context.write(outputKey, outputValue);

    } catch (NumberFormatException e) {

        // Ignore records with invalid salary
    }
}
}

```

Reducer Class

```

public static class Top10Reducer
    extends Reducer<Text, Text, Text, Text> {

    /*
     * Priority queue keeps the 10 highest salaries.
     *
     * Smallest salary stays at the top.
     * When the queue becomes larger than 10,
     * the smallest salary is removed.
     */

    private PriorityQueue<JobRecord> top10;

    @Override
    protected void setup(Context context) {

        top10 = new PriorityQueue<>(
            10,
            (a, b) -> Double.compare(
                a.salary,
                b.salary));
    }

    @Override
    public void reduce(
        Text key,
        Iterable<Text> values,
        Context context)
        throws IOException, InterruptedException {

        double salary = Double.parseDouble(
            key.toString());

        for (Text value : values) {

            JobRecord record =
                new JobRecord(
                    salary,
                    value.toString());

            top10.add(record);

            if (top10.size() > 10) {
                top10.poll();
            }
        }
    }
}

```

```

    }
}

@Override
protected void cleanup(Context context)
    throws IOException, InterruptedException {

    /*
     * Convert PriorityQueue to an array
     * and sort from highest salary to lowest.
     */

    JobRecord[] records =
        top10.toArray(
            new JobRecord[0]);

    java.util.Arrays.sort(
        records,
        (a, b) -> Double.compare(
            b.salary,
            a.salary));

    int rank = 1;

    for (JobRecord record : records) {

        context.write(
            new Text(
                String.valueOf(rank)),
            new Text(
                record.salary
                + "\t"
                + record.details));

        rank++;
    }
}

```

JobRecord Helper Class

```

public static class JobRecord {

    double salary;
    String details;

    JobRecord(
        double salary,
        String details) {

        this.salary = salary;
        this.details = details;
    }
}

```

Driver Class

```

public static void main(String[] args)
    throws Exception {

    if (args.length != 2) {

```

```

        System.err.println(
            "Usage: Top10HighestPaidJobs <input> <output>");
    }

    System.exit(-1);
}

Configuration configuration =
    new Configuration();

Job job = Job.getInstance(
    configuration,
    "Top 10 Highest-Paid Individual Jobs");

job.setJarByClass(
    Top10HighestPaidJobs.class);

// Mapper
job.setMapperClass(
    SalaryMapper.class);

// Reducer
job.setReducerClass(
    Top10Reducer.class);

// Mapper output
job.setMapOutputKeyClass(
    Text.class);

job.setMapOutputValueClass(
    Text.class);

// Final reducer output
job.setOutputKeyClass(
    Text.class);

job.setOutputValueClass(
    Text.class);

// Input
FileInputFormat.addInputPath(
    job,
    new Path(args[0]));

// Output
FileOutputFormat.setOutputPath(
    job,
    new Path(args[1]));

System.exit(
    job.waitForCompletion(true)
        ? 0
        : 1);
}
}

```

Execution

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\Top10HighestPaidJobs>hadoop jar Top10HighestPaidJobs.jar Top10HighestPaidJobs /AI_Impact_Jobs/ai_jobs_salaries.csv /AI_Impact_Jobs/mapreduce/top10_highest_paid_jobs

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\Top10HighestPaidJobs>set DEFAULT_LIBEXEC_DIR=C:\hadoop\bin\..\libexec

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\Top10HighestPaidJobs>if not defined HADOOP_LIBEXEC_DIR (set HADOOP_LIBEXEC_DIR=C:\hadoop\bin\..\libexec )

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\Top10HighestPaidJobs>call C:\hadoop\bin\..\libexec\hadoop-config.cmd jar Top10HighestPaidJobs.jar Top10HighestPaidJobs /AI_Impact_Jobs/ai_jobs_salaries.csv /AI_Impact_Jobs/mapreduce/top10_highest_paid_jobs
JAVA_HOME=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10\bin\java.exe]
CHECKING=[C:\PROGRAM-1\ECLIPSE-1\JDK-17-1.10\bin\java.exe]
2026-09-12 21:07:22,601 INFO client.DefaultHadoopShellOverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 21:07:22,895 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Walid/.staging/job_1789219602256_0021
2026-09-12 21:07:23,101 INFO Input.FileInputFormat: Total input files to process : 1
2026-09-12 21:07:23,160 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-12 21:07:23,270 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789219602256_0021
2026-09-12 21:07:23,270 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-12 21:07:23,416 INFO conf.Configuration: resource-types.xml not found
2026-09-12 21:07:23,417 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-12 21:07:23,478 INFO impl.YarnClientImpl: Submitted application application_1789219602256_0021
2026-09-12 21:07:23,514 INFO mapreduce.Job: The url to track the job: http://scwdussh:8088/proxy/application_1789219602256_0021/
2026-09-12 21:07:23,515 INFO mapreduce.Job: Running job: job_1789219602256_0021
2026-09-12 21:07:36,706 INFO mapreduce.Job: Job job_1789219602256_0021 running in uber mode : false
2026-09-12 21:07:36,709 INFO mapreduce.Job: map 0% reduce 0%
2026-09-12 21:07:47,884 INFO mapreduce.Job: map 100% reduce 0%
2026-09-12 21:07:50,965 INFO mapreduce.Job: map 100% reduce 100%
2026-09-12 21:08:04,923 INFO mapreduce.Job: Job job_1789219602256_0021 completed successfully
2026-09-12 21:08:04,994 INFO mapreduce.Job: Counters: 54

File System Counters
  FILE: Number of bytes read=3545879
  FILE: Number of bytes written=7729863
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  HDFS: Number of bytes read=9168297
  HDFS: Number of bytes written=500
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0

Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=9165
  Total time spent by all reduces in occupied slots (ms)=8847
  Total time spent by all map tasks (ms)=9165
  Total time spent by all reduce tasks (ms)=8847
  Total vcore-milliseconds taken by all map tasks=9165
  Total vcore-milliseconds taken by all reduce tasks=8847
  Total megabyte-milliseconds taken by all map tasks=9384960
  Total megabyte-milliseconds taken by all reduce tasks=9059328
```

Job submission and execution

```
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0

Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=9165
  Total time spent by all reduces in occupied slots (ms)=8847
  Total time spent by all map tasks (ms)=9165
  Total time spent by all reduce tasks (ms)=8847
  Total vcore-milliseconds taken by all map tasks=9165
  Total vcore-milliseconds taken by all reduce tasks=8847
  Total megabyte-milliseconds taken by all map tasks=9384960
  Total megabyte-milliseconds taken by all reduce tasks=9059328

Map-Reduce Framework
  Map input records=71914
  Map output records=71913
  Map output bytes=3402047
  Map output materialized bytes=3545879
  Input split bytes=122
  Combine input records=0
  Combine output records=0
  Reduce input groups=13580
  Reduce shuffle bytes=3545879
  Reduce input records=71913
  Reduce output records=10
  Spilled Records=143826
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=43
  CPU time spent (ms)=5436
  Physical memory (bytes) snapshot=633094912
  Virtual memory (bytes) snapshot=724527480
  Total committed heap usage (bytes)=317718528
  Peak Map Physical memory (bytes)=400707584
  Peak Map Virtual memory (bytes)=441212928
  Peak Reduce Physical memory (bytes)=252187328
  Peak Reduce Virtual memory (bytes)=293314560

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=9168175
  File Output Format Counters
  Bytes Written=500
```

Job completed successfully

Output

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\Top10HighestPaidJobs>hdfs dfs -ls /AI_Impact_Jobs/mapreduce/top10_highest_paid_jobs
JAVA_HOME=[C:\PROGRA~1\ECLIPS~1\JDK-17~1.10-]
CHECKING=[C:\PROGRA~1\ECLIPS~1\JDK-17~1.10-\bin\java.exe]
Found 2 items
-rw-r--r-- 1 Walid supergroup 0 2026-09-12 21:07 /AI_Impact_Jobs/mapreduce/top10_highest_paid_jobs/_SUCCESS
-rw-r--r-- 1 Walid supergroup 500 2026-09-12 21:07 /AI_Impact_Jobs/mapreduce/top10_highest_paid_jobs/part-r-00000

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\Top10HighestPaidJobs>hdfs dfs -cat /AI_Impact_Jobs/mapreduce/top10_highest_paid_jobs/part-r-00000
JAVA_HOME=[C:\PROGRA~1\ECLIPS~1\JDK-17~1.10-]
CHECKING=[C:\PROGRA~1\ECLIPS~1\JDK-17~1.10-\bin\java.exe]
1 800000.0 Software Engineer Senior-level US On-site
2 800000.0 Architect Senior-level US On-site
3 800000.0 AI Architect Mid-level CA Remote
4 793136.0 Data Engineer Entry-level AT On-site
5 774000.0 Data Analyst Entry-level MX On-site
6 750000.0 Analytics Engineer Mid-level US Remote
7 750000.0 Analytics Engineer Senior-level US On-site
8 750000.0 Research Engineer Mid-level US Remote
9 750000.0 Machine Learning Scientist Mid-level US On-site
10 750000.0 Data Scientist Senior-level US On-site

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\Top10HighestPaidJobs>
```

Top 10 Highest-Paid Individual Job Records — HDFS output (part-r-00000)

1	800000.0	Software Engineer	Senior-level	US	On-site
2	800000.0	Architect	Senior-level	US	On-site
3	800000.0	AI Architect	Mid-level	CA	Remote
4	793136.0	Data Engineer	Entry-level	AT	On-site
5	774000.0	Data Analyst	Entry-level	MX	On-site
6	750000.0	Analytics Engineer	Mid-level	US	Remote
7	750000.0	Analytics Engineer	Senior-level	US	On-site
8	750000.0	Research Engineer	Mid-level	US	Remote
9	750000.0	Machine Learning Scientist	Mid-level	US	On-site
10	750000.0	Data Scientist	Senior-level	US	On-site

5. Pig AI Jobs and Salaries Data Analysis

Five Apache Pig scripts were written and executed in local mode against the dataset. Each script loads the CSV using PigStorage, applies GROUP / FOREACH / ORDER / LIMIT operations as needed, and stores the result back using PigStorage with a pipe (|) or comma (,) delimiter.

5.1 Top 10 Highest-Paying Job Roles : Returns the Top 10 highest-paying job role families

Groups all records by role_family, calculates the average salary_in_usd per group, sorts the groups in descending order of average salary, and limits the result to the top 10 role families.

Script

```
-- =====
-- PIG OPERATION 1
-- Top 10 Highest-Paying Job Role Families
-- =====

-- Load the AI Jobs and Salaries dataset from HDFS
jobs = LOAD '/AI_Impact_Jobs/ai_jobs_salaries.csv'
USING PigStorage(',')
AS (
  work_year:int,
  experience_level:chararray,
  employment_type:chararray,
  job_title:chararray,
  salary:double,
  salary_currency:chararray,
  salary_in_usd:double,
  employee_residence:chararray,
  remote_ratio:int,
  company_location:chararray,
  company_size:chararray,
  experience_level_label:chararray,
  employment_type_label:chararray,
  work_mode:chararray,
  salary_outlier_flag:chararray,
  role_family:chararray,
  isco_group_hint:chararray
);

-- Group all records by role family
grouped_roles = GROUP jobs BY role_family;

-- Calculate the average salary for each role family
average_salary = FOREACH grouped_roles GENERATE
  group AS role_family,
  AVG(jobs.salary_in_usd) AS average_salary;

-- Sort role families by average salary in descending order
sorted_roles = ORDER average_salary BY average_salary DESC;

-- Select the top 10 highest-paying role families
top_10_roles = LIMIT sorted_roles 10;

-- Store the final result in HDFS
STORE top_10_roles
INTO '/AI_Impact_Jobs/pig/top10_highest_paying_roles'
USING PigStorage(',');
```

Execution

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\PigAnalysis\scripts>pig -x mapreduce Top10HighestPayingRoles.pig
JAVA_HOME=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-]
CHECKING=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-\bin\java.exe]
2026-09-12 19:22:18,595 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-12 19:22:18,595 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig\1789219338892.log
2026-09-12 19:22:19,120 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Walid\pigbootstrap not found
2026-09-12 19:22:19,168 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-12 19:22:19,168 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-12 19:22:19,563 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-Top10HighestPayingRoles.pig-ce9a25f8-9970-49c6-b7f4-bf83ab4a6b6d
2026-09-12 19:22:19,563 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-12 19:22:20,014 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-12 19:22:20,029 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,LIMIT
2026-09-12 19:22:20,056 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-12 19:22:20,082 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByCon
stParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantC
alculator, SplitFilter, StreamTypeCastInserter]}
2026-09-12 19:22:20,128 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, usageThreshold = 734003200
2026-09-12 19:22:20,168 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-12 19:22:20,201 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-12 19:22:20,223 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizer - Using Secondary Key Optimization for MapReduce node scope-70
2026-09-12 19:22:20,230 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-12 19:22:20,230 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-12 19:22:20,299 [main] INFO org.apache.hadoop.yarn.client.DefaultHARFallowerProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 19:22:20,454 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-12 19:22:20,458 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-12 19:22:20,458 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-12 19:22:20,459 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-12 19:22:20,461 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-12 19:22:20,463 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-12 19:22:20,463 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputS
izeReducerEstimator
2026-09-12 19:22:20,465 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=9168175
2026-09-12 19:22:20,465 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-12 19:22:20,465 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Pig script settings are added to the job
2026-09-12 19:22:20,465 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-12 19:22:20,481 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-12 19:22:20,646 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp118788498
7/tmp-1469301724/pig-0.18.0-core-h3.jar
2026-09-12 19:22:20,646 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-12 19:22:20,669 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp1187884
987/tmp903821991/automaton-1.11-8.jar
2026-09-12 19:22:20,669 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-12 19:22:20,669 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp118788
4087/tmp1156383126/antlr-runtime-3.4.jar
2026-09-12 19:22:20,686 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
```

Script execution

```
HadoopVersion PigVersion UserId StartedAt FinishedAt Features
3.5.0 0.18.0 Walid 2026-09-12 19:32:44 2026-09-12 19:36:14 GROUP_BY,ORDER_BY,LIMIT

Success!

Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReductime Alias Feature Outputs
job_1789219602256_0001 1 1 11 11 11 11 10 10 10 10 average_salary,grouped_roles,jobs GROUP_BY,COMBINER
job_1789219602256_0002 1 1 10 10 10 10 10 10 10 10 sorted_roles SAMPLER
job_1789219602256_0003 1 1 10 10 10 10 10 10 10 10 sorted_roles ORDER_BY,COMBINER
job_1789219602256_0004 1 1 10 10 10 10 10 10 10 10 sorted_roles /AI_Impact_Jobs/pig/top10_highest_paying_roles,

Input(s):
Successfully read 71914 records (9168559 bytes) from: "/AI_Impact_Jobs/ai_jobs_salaries.csv"

Output(s):
Successfully stored 10 records (332 bytes) in: "/AI_Impact_Jobs/pig/top10_highest_paying_roles"

Counters:
Total records written : 10
Total bytes written : 332
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789219602256_0001 -> job_1789219602256_0002,
job_1789219602256_0002 -> job_1789219602256_0003,
job_1789219602256_0003 -> job_1789219602256_0004,
job_1789219602256_0004
```

Successful completion

Output

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\PigAnalysis\scripts>hdfs dfs -ls /AI_Impact_Jobs/pig
JAVA_HOME=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-]
CHECKING=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-\bin\java.exe]
Found 1 items
drwxr-xr-x - Walid supergroup 0 2026-09-12 19:36 /AI_Impact_Jobs/pig/top10_highest_paying_roles

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\PigAnalysis\scripts>hdfs dfs -cat /AI_Impact_Jobs/pig/top10_highest_paying_roles/part-r-00000
JAVA_HOME=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-]
CHECKING=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-\bin\java.exe]
AI Architect,212067.96803652967
Research Scientist,195300.18267882188
ML Engineer,187824.77512355847
Analytics Manager,185237.58585858587
Computer Vision,170379.00714285715
AI Engineer,163843.3399503722
Other / Unclassified,153403.35735076468
Data Scientist,151556.63511640797
Data Engineer,144765.3288711013
NLP,134649.15789473685
```

Top 10 Highest-Paying Job Roles — output

5.2 Salary Analysis by Experience Level : Returns average, maximum and minimum salary for each experience level

Groups all records by experience_level_label and calculates the average, maximum, and minimum salary_in_usd for each group, sorted from highest to lowest average salary.

Script

```
-- =====
-- PIG OPERATION 2
-- Salary Analysis by Experience Level
-- =====

jobs = LOAD '/AI_Impact_Jobs/ai_jobs_salaries.csv'
USING PigStorage(',')
AS (
  work_year:int,
  experience_level:chararray,
  employment_type:chararray,
  job_title:chararray,
  salary:double,
  salary_currency:chararray,
  salary_in_usd:double,
  employee_residence:chararray,
  remote_ratio:int,
  company_location:chararray,
  company_size:chararray,
  experience_level_label:chararray,
  employment_type_label:chararray,
  work_mode:chararray,
  salary_outlier_flag:chararray,
  role_family:chararray,
  isco_group_hint:chararray
);

-- Remove CSV header
jobs = FILTER jobs BY work_year IS NOT NULL;

-- Group jobs by experience level
grouped_experience = GROUP jobs BY experience_level_label;

-- Calculate salary statistics
salary_analysis = FOREACH grouped_experience GENERATE
  group AS experience_level,
  AVG(jobs.salary_in_usd) AS average_salary,
  MAX(jobs.salary_in_usd) AS maximum_salary,
  MIN(jobs.salary_in_usd) AS minimum_salary;

-- Sort by average salary from highest to lowest
sorted_experience = ORDER salary_analysis BY average_salary DESC;

-- Store result in HDFS
STORE sorted_experience
INTO '/AI_Impact_Jobs/pig/salary_by_experience'
USING PigStorage(',');
```

Script execution

Successful completion

```
Executive-level,202080.47,625000.0,15000.0
Senior-level,167807.96,800000.0,15645.0
Mid-level,136267.01,800000.0,15000.0
Entry-level,98017.08,793136.0,15000.0
```

Salary Analysis by Experience Level — output

5.3 Job Distribution by Work Mode : Returns job count and percentage share for each work mode

Groups all records by work_mode, counts the jobs per group, and cross-joins against the overall total job count to calculate each work mode's percentage share, sorted by job count.

Script

```
-- =====
-- PIG OPERATION 3
-- Job Distribution by Work Mode
-- =====

jobs = LOAD '/AI_Impact_Jobs/ai_jobs_salaries.csv'
USING PigStorage(',')
AS (
  work_year:int,
  experience_level:chararray,
  employment_type:chararray,
  job_title:chararray,
  salary:double,
  salary_currency:chararray,
  salary_in_usd:double,
  employee_residence:chararray,
  remote_ratio:int,
  company_location:chararray,
  company_size:chararray,
  experience_level_label:chararray,
  employment_type_label:chararray,
  work_mode:chararray,
  salary_outlier_flag:chararray,
  role_family:chararray,
  isco_group_hint:chararray
);

jobs = FILTER jobs BY work_year IS NOT NULL;

grouped_work_mode = GROUP jobs BY work_mode;

work_mode_count = FOREACH grouped_work_mode GENERATE
  group AS work_mode,
  COUNT(jobs) AS job_count;

total_jobs_group = GROUP jobs ALL;

total_jobs = FOREACH total_jobs_group GENERATE
  COUNT(jobs) AS total_count;

crossed_data = CROSS work_mode_count, total_jobs;

work_mode_percentage = FOREACH crossed_data GENERATE
  work_mode,
  job_count,
  (job_count * 100.0 / total_count) AS percentage;

sorted_work_mode = ORDER work_mode_percentage BY job_count DESC;

STORE sorted_work_mode
INTO '/AI_Impact_Jobs/pig/jobs_by_work_mode'
USING PigStorage(',');
```

Execution

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\PigAnalysis\scripts>pig -x mapreduce jobsByWorkMode.pig
JAVA_HOME=[C:\PROGRAM-1\ECLIPS-1\JDK-17-1.10-]
CHECKING=[C:\PROGRAM-1\ECLIPS-1\JDK-17-1.10-\bin\java.exe]
2026-09-12 19:56:35,680 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-12 19:56:35,680 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-12 19:56:35,680 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-12 19:56:36,024 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-12 19:56:36,024 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789221396010.log
2026-09-12 19:56:36,325 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Walid/.pigbootstrap not found
2026-09-12 19:56:36,387 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-12 19:56:36,387 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-12 19:56:36,832 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-JobsByWorkMode.pig-81b0494b-a62a-44d6-9d63-670751568443
2026-09-12 19:56:36,832 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-12 19:56:37,350 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning DPLCIT(CAST TO DOUBLE 2 time(s).
2026-09-12 19:56:37,354 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-12 19:56:37,367 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER,CROSS
2026-09-12 19:56:37,414 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-12 19:56:37,430 [main] INFO org.apache.pig.newplan.LogicalOptimizer.LogicalPlanOptimizer - (RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByCon
ParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushdownForEachFlatten, PushMapFilter, SplitConstant
Calculator, SplitFilter, StreamTypeCastInserter])
2026-09-12 19:56:37,481 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor: collectionUsageThreshold = 734803200, usageThreshold = 734803200
2026-09-12 19:56:37,541 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-12 19:56:37,557 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-12 19:56:37,573 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-12 19:56:37,581 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-134
2026-09-12 19:56:37,589 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MRCompiler$LastInputStreamingOptimizer - Rewrite: POPackage->POForEach to POPackage(JoinPacker)
2026-09-12 19:56:37,595 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MRCompiler - MR plan size before optimization: 6
2026-09-12 19:56:37,595 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - Merged MR job 121 into MR job 118
2026-09-12 19:56:37,595 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - Merged MR job 126 into MR job 118
2026-09-12 19:56:37,595 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - Requested parallelism of splitter: 1
2026-09-12 19:56:37,595 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - Merged 2 map-reduce splittes.
2026-09-12 19:56:37,595 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - Merged 2 out of total 3 MR operators.
2026-09-12 19:56:37,605 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-12 19:56:37,668 [main] INFO org.apache.hadoop.yarn.client.DefaultHadoopWailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 19:56:37,811 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-12 19:56:37,811 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-12 19:56:37,819 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-12 19:56:37,819 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-12 19:56:37,819 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-12 19:56:37,819 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - Setting Parallelism to 1
2026-09-12 19:56:37,819 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reducers
2026-09-12 19:56:37,819 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - This job cannot be converted run in-process
2026-09-12 19:56:37,835 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-12 19:56:38,005 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - Added jar file:C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp-17898803
o/tmp-06629315/pig-0.18.0-core-h3.jar
2026-09-12 19:56:38,005 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-12 19:56:38,030 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp-178998
```

Script execution

```
HadoopVersion PigVersion UserId StartedAt FinishedAt Features
3.5.0 0.18.0 Walid 2026-09-12 19:56:37 2026-09-12 20:00:19 GROUP_BY,ORDER_BY,FILTER,CROSS

Success!

Job Stats (time in seconds):
JobId Maps JobId Maps MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReduceTime Alias Feature Outputs
job_1789219602256_0008 1 1 11 11 11 10 10 10 10 grouped_work_mode,jobs,total_jobs,total_jobs_group,work_mode_count MULTI_QUERY,COMBINER
job_1789219602256_0009 2 1 10 10 10 23 23 23 23 crossed_data,work_mode_percentage
job_1789219602256_0010 1 1 10 10 10 10 10 10 10 sorted_work_mode SAMPLER
job_1789219602256_0011 1 1 10 10 10 10 10 10 10 sorted_work_mode ORDER_BY /AI_Impact_Jobs/pig/jobs_by_work_mode,

Input(s):
Successfully read 71914 records (916859 bytes) from: "/AI_Impact_Jobs/ai_jobs_salaries.csv"

Output(s):
Successfully stored 3 records (94 bytes) in: "/AI_Impact_Jobs/pig/jobs_by_work_mode"

Counters:
Total records written : 3
Total bytes written : 94
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789219602256_0008 -> job_1789219602256_0009,
job_1789219602256_0009 -> job_1789219602256_0010,
job_1789219602256_0010 -> job_1789219602256_0011,
job_1789219602256_0011
```

Successful completion

Output

On-site,54081,75.20%
Remote,17505,24.34%
Hybrid,327,0.45%

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\PigAnalysis\scripts>hdfs dfs -cat /AI_Impact_Jobs/pig/jobs_by_work_mode/part-r-00000
JAVA_HOME=[C:\PROGRAM-1\ECLIPS-1\JDK-17-1.10-]
CHECKING=[C:\PROGRAM-1\ECLIPS-1\JDK-17-1.10-\bin\java.exe]
On-site,54081,75.20337073964373
Remote,17505,24.34191314505027
Hybrid,327,0.45471611538599476

C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\PigAnalysis\scripts>
```

Job Distribution by Work Mode — output

5.4 Top 10 Job Locations : Returns the top 10 company locations by number of AI jobs

Groups all records by company_location, counts the number of jobs per location, sorts the groups in descending order of job count, and limits the result to the top 10 locations.

Script

```
-- =====
-- PIG OPERATION 4
-- Top 10 Company Locations by Number of AI Jobs
-- =====

jobs = LOAD '/AI_Impact_Jobs/ai_jobs_salaries.csv'
USING PigStorage(',')
AS (
    work_year:int,
    experience_level:chararray,
    employment_type:chararray,
    job_title:chararray,
    salary:double,
    salary_currency:chararray,
    salary_in_usd:double,
    employee_residence:chararray,
    remote_ratio:int,
    company_location:chararray,
    company_size:chararray,
    experience_level_label:chararray,
    employment_type_label:chararray,
    work_mode:chararray,
    salary_outlier_flag:chararray,
    role_family:chararray,
    isco_group_hint:chararray
);

jobs = FILTER jobs BY work_year IS NOT NULL;

grouped_locations = GROUP jobs BY company_location;

location_count = FOREACH grouped_locations GENERATE
    group AS company_location,
    COUNT(jobs) AS job_count;

sorted_locations = ORDER location_count BY job_count DESC;

top_10_locations = LIMIT sorted_locations 10;

STORE top_10_locations
INTO '/AI_Impact_Jobs/pig/top10_job_locations'
USING PigStorage(',');
```

Script execution

Successful completion

Output

Top 10 Job Locations — output

5.5 Salary Comparison by Employment Type : Returns average, maximum and minimum salary for each employment type

Groups all records by employment_type_label and calculates the average, maximum, and minimum salary_in_usd for each group, sorted by average salary in descending order.

Script

```
-- =====
-- PIG OPERATION 5
-- Salary Comparison by Employment Type
-- =====

jobs = LOAD '/AI_Impact_Jobs/ai_jobs_salaries.csv'
USING PigStorage(',')
AS (
  work_year:int,
  experience_level:chararray,
  employment_type:chararray,
  job_title:chararray,
  salary:double,
  salary_currency:chararray,
  salary_in_usd:double,
  employee_residence:chararray,
  remote_ratio:int,
  company_location:chararray,
  company_size:chararray,
  experience_level_label:chararray,
  employment_type_label:chararray,
  work_mode:chararray,
  salary_outlier_flag:chararray,
  role_family:chararray,
  isco_group_hint:chararray
);

jobs = FILTER jobs BY work_year IS NOT NULL;

grouped_employment = GROUP jobs BY employment_type_label;

salary_comparison = FOREACH grouped_employment GENERATE
  group AS employment_type,
  AVG(jobs.salary_in_usd) AS average_salary,
  MAX(jobs.salary_in_usd) AS maximum_salary,
  MIN(jobs.salary_in_usd) AS minimum_salary;

sorted_employment = ORDER salary_comparison BY average_salary DESC;

STORE sorted_employment
INTO '/AI_Impact_Jobs/pig/salary_by_employment_type'
USING PigStorage(',');
```

Execution

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\PigAnalysis\scripts>pig -x mapreduce SalaryByEmploymentType.pig
JAVA_HOME=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-]
CHECKING=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-\bin\java.exe]
2026-09-12 20:11:15,338 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-12 20:11:15,340 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-12 20:11:15,340 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-12 20:11:15,677 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-12 20:11:15,677 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789222395670.log
2026-09-12 20:11:15,973 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Walid\pig\bootstrap not found
2026-09-12 20:11:16,028 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-12 20:11:16,032 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-12 20:11:16,478 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-SalaryByEmploymentType.pig-6e2f769c-8d33-4f6d-89b9-52dc01b7f306
2026-09-12 20:11:16,478 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-12 20:11:17,085 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-12 20:11:17,036 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schemaTuple] was not set... will not generate code.
2026-09-12 20:11:17,071 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - (RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByCon
stParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantC
alculator, SplitFilter, StreamTypeCastInserter])
2026-09-12 20:11:17,113 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734803200, usageThreshold = 734003200
2026-09-12 20:11:17,160 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-12 20:11:17,210 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-12 20:11:17,231 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.SecondaryKeyOptimizerRW - Using Secondary Key Optimization for MapReduce node scope-82
2026-09-12 20:11:17,245 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-12 20:11:17,245 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-12 20:11:17,324 [main] INFO org.apache.hadoop.yarn.client.DefaultHARFollowerProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-12 20:11:17,458 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-12 20:11:17,464 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-12 20:11:17,465 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-12 20:11:17,465 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-12 20:11:17,467 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-12 20:11:17,469 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-12 20:11:17,471 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.InputS
izeReducerEstimator
2026-09-12 20:11:17,478 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=9168175
2026-09-12 20:11:17,478 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-12 20:11:17,478 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-12 20:11:17,478 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/tmp-10368045
61/tmp-63909695/pig-0.18.0-core-h3.jar
2026-09-12 20:11:17,781 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/tmp-103680
4561/tmp-217645220/automaton-1.11-8.jar
2026-09-12 20:11:17,730 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/tmp-103680
4561/tmp-217645220/automaton-1.11-8.jar
2026-09-12 20:11:17,730 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/tmp-103680
4561/tmp-217645220/automaton-1.11-8.jar
2026-09-12 20:11:17,754 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/tmp-103680
4561/tmp-217645220/automaton-1.11-8.jar
2026-09-12 20:11:17,754 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file:C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/tmp-103680
4561/tmp-217645220/automaton-1.11-8.jar
```

Script execution

```
HadoopVersion PigVersion UserId StartedAt FinishedAt Features
3.5.0 0.18.0 Walid 2026-09-12 20:13:17 2026-09-12 20:15:51 GROUP_BY,ORDER_BY,FILTER

Success!

Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReduceTime Alias Feature Outputs
job_1789219602256_0016 1 1 11 11 11 11 10 10 10 10 grouped_employment,jobs,salary_comparison GROUP_BY,COMBINER
job_1789219602256_0017 1 1 10 10 10 10 10 10 10 10 sorted_employment SAMPLER
job_1789219602256_0018 1 1 10 10 10 10 10 10 10 10 sorted_employment ORDER_BY /AI_Impact_Jobs/pig/salary_by_employment_type,

Input(s):
Successfully read 71914 records (9168559 bytes) from: "/AI_Impact_Jobs/ai_jobs_salaries.csv"

Output(s):
Successfully stored 4 records (173 bytes) in: "/AI_Impact_Jobs/pig/salary_by_employment_type"

Counters:
Total records written : 4
Total bytes written : 173
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789219602256_0016 -> job_1789219602256_0017,
job_1789219602256_0017 -> job_1789219602256_0018,
job_1789219602256_0018
```

Successful completion

Output

```
Full-time,151785.78,800000.0,15000.0
Contract,104232.66,500000.0,16000.0
Part-time,80330.19,405000.0,15966.0
Freelance,50651.56,100000.0,20000.0
```

```
C:\Users\Walid\Desktop\AI-Impact-on-Jobs-and-Salaries\PigAnalysis\scripts>hdfs dfs -cat /AI_Impact_Jobs/pig/salary_by_employment_type/part-r-000000
JAVA_HOME=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-]
CHECKING=[C:\PROGRA~1\ECLIPS~1\JDK-17-1.10-\bin\java.exe]
Full-time,151785.7837359302,800000.0,15000.0
Contract,104232.66057441253,500000.0,16000.0
Part-time,80330.19373219373,405000.0,15966.0
Freelance,50651.5625,100000.0,20000.0
```

Salary Comparison by Employment Type — output